

ELARA-U: A Cross-Domain Anomaly
Meta-Routing Benchmark
—Stacking Beats Selection; Reliability Routing
Helps Only Under Independent Modality Failure

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Thesis statement. The central problem in universal anomaly detection is not only inventing a stronger detector, but preventing the wrong detector from being trusted under domain shift. ELARA-U studies this as a meta-routing problem: given a heterogeneous detector zoo and a validation slice, decide when to select, fuse, stack, or fall back without using test labels. On the clean i.i.d. archive of 123 tasks from 5 families, ELARA-U Stack (per-task rank-normalized stacking) reaches mean rank 2.53 versus 3.38 for validation auto-selection (+0.036 AUROC, paired-bootstrap CI [+0.023, +0.050], excluding zero; 86/123 wins) and even exceeds the best-single-detector oracle (0.784 vs 0.756). However, controlled ablations across four regimes (Holm-corrected) and a *sealed natural temporal-shift* benchmark (D22) resolve RQ3 negatively: reliability/drift features add no significant value in any regime, and a reliability gate on the stack is significantly *worse* (-0.037 AUROC, $p_{\text{Holm}} < 10^{-8}$); under natural shift the drift router is directionally worse (-0.050 AUROC, raw CI excl. 0, but $n=7$ does not survive Holm). The honest contribution is a cross-domain benchmark with a strong stacking baseline and a Holm-robust negative result for reliability-aware routing—not a sealed universal SOTA claim. **The decisive update:** in true multimodal detection with independent modality failure (pre-registered D23, validated on *two* real datasets — Real-IAD-D3 (15 categories) and MVTec-3D (7 categories)), reliability routing wins (beats equal-weight +0.24/+0.21, stale-select +0.39/+0.32, validation-only fusion +0.25/+0.21; all CIs exclude zero on both), while remaining inert when no modality fails. This win is, however, *failure-model-specific*: under a realistic stuck-sensor dropout and under real natural degradation the gate beats stale selection but not plain fusion (n.s.), so its value needs a detectable drift signal. The precise claim is therefore *reliability routing helps iff a modality fails independently and detectably*; for single-input meta-routing plain stacking suffices.

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1 Executive Summary

Anomaly detection is fragmented across tabular, cyber, fraud, image-OOD, text, industrial vision, time-series, and multimodal settings. Each family has strong specialists, but none is uniformly best. In deployment, the damaging failure is often not the absence of a useful expert; it is selecting a detector whose validation performance no longer predicts deployment performance. This is the negative-transfer problem.

ELARA-U reframes the project around a universal anomaly *meta-router*. Instead of claiming that one detector is universal, ELARA-U asks whether a router can reduce average rank, regret to the per-task oracle, and harmful transfer events across many task families. The project therefore moves from the old clean-transfer-only framing to a broader Gate U contract: success is judged by cross-family rank, regret, calibration, negative-transfer reduction, and audited no-test-label routing.

1.1 Current Evidence Level

The current thesis evidence should be read in three tiers.

Table 1: Evidence status of the current ELARA-U thesis.

Evidence tier	What it shows	Claim status
Clean 123-task benchmark	Rank-normalized stacking beats validation auto-select (+0.036 AUROC, CI>0) and the best fixed detector (+0.075) across 5 families.	Established (verified)
Reliability ablation	Reliability/drift features add no significant gain over validation quality; a reliability gate hurts.	Established (negative)
Mechanism (simulation)	Reliability gating helps iff a channel fails independently.	Established (simulation)
Sealed + independent external	Stacking replicates on a sealed ADBench-extractor holdout (D24) and a fully independent corpus (D27, +0.111); reliability wins under multimodal failure (D23).	Done (D23/D24/D27)

The immediate research value is that the work now has a defensible thesis: *meta-routing is most useful when validation metrics become stale under deployment shift*. This is stronger and cleaner than claiming one fixed detector or one clean-transfer gate should win everywhere.

2 Research Questions and Contributions

2.1 Research Questions

Table 2: Core research questions for the ELARA-U thesis.

ID	Question	Current status
RQ1	Can a cross-domain router outperform the best fixed detector family over many anomaly tasks?	Yes (clean 123-task benchmark, CI>0)
RQ2	Can the router (stack) outperform validation-AUROC auto-selection?	Yes (+0.036 AUROC, CI>0)
RQ3	Does reliability/drift modeling, rather than plain stacking, carry the improvement?	NO single-input; YES multimodal (3 single-input regimes vs D23, Table 33)
RQ4	Can the same contract hold on natural, sealed, and external benchmarks (incl. multimodal)?	Largely yes: sealed (D24), fully independent (D27), multimodal (D23); natural shift (D22) mixed
RQ5	Can the router produce calibrated probabilities and safe abstentions suitable for operations?	Partially supported; calibration pending

2.2 Contributions

Table 3: Contribution matrix.

Contribution	Description	Evidence
Universal meta-routing formulation	Defines anomaly detection as a task-level routing problem over a detector zoo, not as a single-detector competition.	Formal setup and Gate U
Score archive	Persists validation/test detector scores, labels, and provenance so routing policies can be evaluated without re-running detectors.	123 current tasks
Rank-normalized stacker	Learns per-task combinations from validation labels and applies monotone-invariant rank features to test scores.	Mean rank 2.53 (clean benchmark)
Evidence contract	Separates average rank, regret, negative transfer, calibration, and no-test-label audit.	Tables and algorithms
Theory stack	Proves why universal dominance is impossible, why auto-select is near-oracle under stable transfer, and why shifted regimes can reward reliability routing.	Theorems in Sec. 8
Honest boundary	Distinguishes development stress evidence from sealed universal SOTA evidence.	Validity audit

3 Problem Definition

Let $\mathcal{T}$ be a collection of anomaly-detection tasks. Each task $\tau \in \mathcal{T}$ has a training split, a validation split, and a deployment or test split. A detector zoo

$$\mathcal{D} = \{d_1, \dots, d_M\}$$

produces detector scores $s_{\tau,m}(x)$ for every sample x . The objective is not to find a detector that wins every task. The objective is to learn or define a router R that maps task evidence into a selected, fused, or stacked score

$$g_\tau(x) = R(\{s_{\tau,m}\}_{m=1}^M, \Phi_\tau)(x),$$

where Φ_τ contains validation-only and unlabeled deployment-shift features. The router is evaluated using test labels only after routing decisions are frozen.

3.1 Metrics

Table 4: Primary metrics used by the thesis.

Metric	Definition	Preferred direction
Mean rank	Average rank of each method by task AUROC.	Lower
Regret	Per-task oracle AUROC minus method AUROC.	Lower
Negative-transfer rate	Fraction of tasks where the method underperforms a safe reference such as rank-mean.	Lower
Bootstrap CI	Paired task-level confidence interval for rank delta.	Exclude zero
ECE proxy	Calibration error treating normalized scores as probabilities.	Lower
Worst rank/regret	Worst observed task-level failure.	Lower

4 Related Work

Benchmarks and the no-free-lunch result. A consistent finding across large anomaly-detection studies is that *no single detector dominates*: ADBench [8] evaluates 30 algorithms over 57 tabular, vision, and language feature sets and reports that the best method varies by dataset; PyOD [9] standardises the detector implementations we use. This per-dataset variability is the empirical premise of our work — it is exactly what makes a routing/combination layer plausible — and our 123-task suite is built from ADBench-style feature vectors across five families, extended here with time-series (NAB, SMD) and a sealed representation-shift holdout.

Model and detector selection. The natural response to no-free-lunch is to *select* the right detector per dataset. MetaOD [7] learns this mapping from dataset meta-features via collaborative filtering; classical AutoML selectors and landmarker-based meta-learning follow the same template. The strong label-aware analogue is validation-AUROC auto-selection, which is our primary baseline. A central contribution of this chapter is the (perhaps surprising) negative result that selection — even a learned MetaOD-style meta-selector — is the *wrong primitive*: it does not beat naive auto-select on our suite, and is itself beaten by stacking (Section 19).

Ensembles, fusion, and stacking. An alternative to selection is to *combine* detectors. Score normalisation and averaging, rank aggregation, and greedy ensemble selection are common in outlier-ensemble work; stacked generalisation [10] fits a meta-model over base-learner outputs. We find that a rank-normalised logistic stack is the strongest combiner in this family — it beats selection with confidence intervals excluding zero, is robust to the stacker family (random forest, gradient boosting), and *outperforms* the simple average, which itself loses to selection. The contribution is not a new combiner but a clean, hardened demonstration of *when* combination beats selection across heterogeneous anomaly families.

Distribution shift, drift, and selective prediction. Test-time shift is classically detected with two-sample statistics such as Kolmogorov–Smirnov, and selective prediction / learning-to-abstain [11] motivates a fallback action when no expert is trustworthy. Our *reliability* machinery (validation–test KS drift, calibration, inter-detector disagreement) is built from exactly these signals. Our honest finding is a boundary result: these signals do *not* add value over plain stacking in single-input regimes, and help only when modalities can fail *independently* (Section 20).

Calibration. Beyond ranking quality, deployable detectors need calibrated scores. We report fitted Brier and negative log-likelihood under isotonic-regression calibration [6], rather than a raw ECE proxy.

Multimodal and industrial anomaly detection. Industrial inspection provides genuinely multimodal data (RGB, photometric stereo, point cloud): PatchCore [4] and Real-IAD supply the detectors and co-observed modalities behind our independent-modality-failure study (D23), the one regime where reliability routing wins over plain fusion (and then only under a drift-inducing failure). Deep one-class methods [3] and OOD baselines [5] are included in or motivate our detector zoo and image-OOD family.

Positioning. In short, prior work establishes (i) that no detector is universal, and (ii) two competing remedies — meta-selection and ensembling. This chapter adjudicates between them on a broad, pre-registered, leakage-audited suite and reaches a sharp conclusion: *stacking beats selection; reliability routing helps only under independent modality failure.*

5 From the Old ELARA System to ELARA-U

The prior ELARA/RGA work remains useful, but it should be treated as source-system evidence rather than the main flagship claim. Its value is threefold: it provides a detector-and-fusion foundation, it supplies negative evidence about near-ceiling clean transfer, and it motivates a broader meta-routing benchmark.

Table 5: How old artifacts should be reused in the new thesis.

Old artifact	Reuse in ELARA-U	Do not use as
RGA/RGA+ source system	Detector/fusion foundation and stress mechanism.	Universal proof
Clean-transfer failures	Motivation for the new routing objective.	A hidden pass
T1–T9 theory stack	Boundary theorems explaining when clean transfer is hard.	A reason to overclaim
Real-IAD / Real-IAD D3	Natural-degradation evidence track and future family input.	Fresh sealed evidence if already opened
MVTec 3D-AD / 3D-ADAM / MulSen	Multimodal source families and development evidence.	Final universal holdout without preregistration

This distinction is important. A failed clean-transfer gate does not make the project worthless; it says that the old claim was too narrow. The stronger scientific move is to make the failure part of the motivation: when clean validation-to-test transfer is stable, validation auto-selection is already a strong baseline. A universal paper should therefore focus on the regimes where selection becomes stale.

6 Methodology: The Reliability-Aware Meta-Router

ELARA-U operates in two conceptually distinct modes.

Table 6: ELARA-U operating modes.

Mode	Behavior	Best use
Clean baseline mode	Select the detector with highest validation AUROC.	Stable validation-to-test transfer
Shift-aware stack mode	Fit a regularized rank-normalized stacker on validation data and apply it to test ranks.	Stale validation under score shift or degradation
Fallback mode	Use rank-mean or abstain when channels are degenerate.	Unsafe or untrusted routing

6.1 Detector Zoo and Router Features

Table 7: Detector zoo. Current archive uses the tabular/cyber/fraud/image/text subsets; the remaining families are thesis expansion targets.

Domain	Experts
Tabular	ECOD, COPOD, Isolation Forest, LOF, KNN, OCSVM
Cyber/fraud	One-class and density detectors over transaction or network features
Image-OOD	Feature-space anomaly detectors over CNN representations
Text	Feature-space anomaly detectors over sentence/document embeddings
Time series	Matrix profile, forecasting residual, reconstruction, and sequence models
Industrial vision	PatchCore, reverse-distillation, EfficientAD-style experts
Multimodal/3D	RGB, depth, point-cloud, SAR/CW-style multimodal fusions

Algorithm 1 Universal Score Archive Construction

Require: datasets $\mathcal{D} = \{X_d\}$, detector zoo $\mathcal{E} = \{e_1, \dots, e_M\}$ **Ensure:** score archive $\mathcal{A}$, validation summaries V

```

1: for each dataset  $d \in \mathcal{D}$  do
2:   split  $X_d$  into train/val/test; fit scaler on train
3:   for each expert  $e_m \in \mathcal{E}$  do
4:     fit  $e_m$  on (unlabelled) train;  $s_{d,m}^{\text{val}} \leftarrow e_m(\text{val})$ ,  $s_{d,m}^{\text{test}} \leftarrow e_m(\text{test})$ 
5:     normalise by monotone  $z$ -sigmoid on the train-OK score distribution
6:      $v_{d,m} \leftarrow \text{AUROC}(y_{\text{val}}, s_{d,m}^{\text{val}})$  ▷ validation labels only
7:     store  $(s_{d,m}^{\text{val}}, s_{d,m}^{\text{test}}, v_{d,m})$  and provenance (config hash, artifact hash) in  $\mathcal{A}$ 
8:   end for
9: end for
10: return  $\mathcal{A}$ ,  $V = \{v_{d,m}\}$ 

```

Algorithm 2 Validation-Only Reliability Router Training

Require: archive $\mathcal{A}$, validation summaries V , dataset meta-features Φ **Ensure:** frozen router R

```

1: for each  $(d, m)$  do
2:    $r_{d,m} \leftarrow [\text{score tailness, entropy, inter-expert disagreement, calibration error,}$ 
      $\text{missing-modality flags, domain-shift estimate, } v_{d,m}, \Phi_d]$  ▷ no test labels
3: end for
4: target  $t_d \leftarrow \arg \max_m v_{d,m}$  (or the per-dataset rank vector)
5: fit  $R: r \mapsto (\text{expert choice} / \text{fusion weights})$  on  $\{(r_{d,\cdot}, t_d)\}$  using validation only
6: audit: assert no test split or test label entered  $R$ 's fit
7: return frozen  $R$ 

```

Table 8: Router feature families.

Group	Examples	Labels
Validation reliability	validation AUROC, spread, confidence, inversion flags	validation only
Score geometry	entropy, sharpness, tail mass, saturation, rank dispersion	no labels
Disagreement	inter-expert rank and score disagreement	no labels
Shift	validation-to-test KS drift, score distribution drift	no test labels
Dataset meta-features	sample count, dimension, anomaly rate on validation	validation only
Calibration	ECE proxy, future Brier/NLL after fitted calibration	validation only

7 Theory Stack

The theory stack is not a proof that ELARA-U is universal SOTA. It is a proof of when the claim is possible and when it is structurally impossible. The key message is:

ELARA-U helps when stale-validation regret Δ exceeds router prediction error 2η .

If validation and deployment detector rankings remain stable, auto-selection is already near-oracle. If deployment shift changes detector ranking and the router can predict that change from validation-only and unlabeled shift evidence, routing can reduce regret.

Algorithm 3 Routed Inference With Abstention

Require: frozen router R , calibrator g , abstention threshold τ_a , new task scores $\{s_m\}$ with validation slice

Ensure: calibrated routed score or abstention, with reason code

- 1: $r \leftarrow$ reliability features on the task’s validation slice (Alg. 2)
 - 2: $(w, \hat{\rho}) \leftarrow R(r)$ $\triangleright$ fusion weights w and predicted reliability $\hat{\rho}$
 - 3: **if** $\hat{\rho} < \tau_a$ **or** all experts flagged degenerate **then**
 - 4: **return** ABSTAIN with reason code (low predicted reliability)
 - 5: **end if**
 - 6: $\tilde{s} \leftarrow \sum_m w_m s_m$ over non-degenerate experts (degenerate-channel guard)
 - 7: **return** $g(\tilde{s})$ (calibrated) with reason code (selected experts, weights)
-

Algorithm 4 Universal Evidence Evaluation (Gate U)

Require: per-task test AUROCs for router and baselines over tasks $\mathcal{T}$

Ensure: Gate U evidence report

- 1: **for** each task $\tau \in \mathcal{T}$ **do**
 - 2: rank methods by test AUROC; $\text{oracle}_\tau \leftarrow \max \text{AUROC}$
 - 3: $\text{regret}_\tau(\cdot) \leftarrow \text{oracle}_\tau - \text{AUROC}_\tau(\cdot)$; flag negative transfer vs best fixed
 - 4: **end for**
 - 5: aggregate mean rank, win rate, mean/worst regret, negative-transfer rate
 - 6: paired bootstrap over $\mathcal{T}$ for (router – best-fixed) rank gain $\Rightarrow$ 95% CI
 - 7: Holm-correct across the endpoint family; **return** report
-

8 Theory Stack for ELARA-U: Shift-Aware Reliability Routing

8.1 Notation and Setup

Let $\mathcal{D} = \{1, \dots, M\}$ be a finite detector zoo. For a task τ , detector $m \in \mathcal{D}$ produces an anomaly score $s_m(x) \in \mathbb{R}$. Let V_τ denote the validation distribution and T_τ denote the deployment/test distribution. Let

$$A_V(m) = \text{AUROC}_{V_\tau}(s_m), \quad A_T(m) = \text{AUROC}_{T_\tau}(s_m).$$

The validation auto-selector is

$$m_{\text{val}} = \arg \max_{m \in \mathcal{D}} A_V(m),$$

and the test oracle is

$$m^* = \arg \max_{m \in \mathcal{D}} A_T(m).$$

The test regret of a method h is

$$\text{Reg}_T(h) = A_T(m^*) - A_T(h).$$

When h outputs a fused or stacked score $g_h(x)$ instead of a single detector, define

$$A_T(h) = \text{AUROC}_{T_\tau}(g_h).$$

ELARA-U observes validation-only reliability features and unlabeled deployment-shift features, including score dispersion, entropy, inter-detector disagreement, calibration error, detector degeneracy flags, and train-validation/test-score drift statistics. It never observes test labels during routing.

8.2 Main Evidentiary Theorems

Theorem 1 (No Universal Dominance Without Assumptions). *For any deterministic detector, fusion rule, or meta-router h that selects or combines scores from a finite detector zoo $\mathcal{D}$, there exists a pair of anomaly-detection tasks (τ_1, τ_2) such that h cannot be strictly optimal on both tasks. Therefore, no universal anomaly detector or router can dominate all alternatives across all possible domains without assumptions on task structure, reliability, or shift.*

Proof. Fix any method h . Since h is deterministic, for a given task representation it outputs either one detector or a fixed scoring function g_h built from detector scores.

Construct a task τ_1 where g_h ranks all anomalous examples above all normal examples. Then

$$A_{\tau_1}(h) = 1.$$

Now construct a second task τ_2 on the same score space by reversing the anomaly labels: all examples ranked high by g_h are normal and all examples ranked low by g_h are anomalous. Under this label reversal,

$$A_{\tau_2}(h) = 0.$$

However, there exists another detector m' that ranks the τ_2 anomalies above normals, achieving

$$A_{\tau_2}(m') = 1.$$

□

Thus h is optimal on τ_1 but maximally suboptimal on τ_2 . Since this construction can be made for any fixed h , no detector, fusion rule, or router can be universally dominant without additional assumptions.

Interpretation. This theorem justifies the ELARA-U claim structure: the paper does not claim “best on every dataset.” The defensible claim is reduction of regret and negative transfer under specified reliability and shift conditions.

Theorem 2 (Validation Auto-Selection Is Near-Oracle Under Stable Transfer). *Assume validation and deployment detector performances are uniformly close:*

$$\sup_{m \in \mathcal{D}} |A_T(m) - A_V(m)| \leq \varepsilon.$$

Then validation-AUROC auto-selection has deployment regret bounded by

$$\text{Reg}_T(m_{\text{val}}) \leq 2\varepsilon.$$

Proof. Let $m^* = \arg \max_m A_T(m)$ be the test oracle. By definition of m_{val} ,

$$A_V(m_{\text{val}}) \geq A_V(m^*).$$

Using the uniform stability assumption,

$$A_T(m_{\text{val}}) \geq A_V(m_{\text{val}}) - \varepsilon,$$

and

$$A_V(m^*) \geq A_T(m^*) - \varepsilon.$$

Combining these inequalities gives

$$A_T(m_{\text{val}}) \geq A_V(m_{\text{val}}) - \varepsilon \geq A_V(m^*) - \varepsilon \geq A_T(m^*) - 2\varepsilon.$$

Therefore,

$$\text{Reg}_T(m_{\text{val}}) = A_T(m^*) - A_T(m_{\text{val}}) \leq 2\varepsilon.$$

□

Interpretation. This theorem explains why validation-AUROC auto-selection is hard to beat on clean i.i.d. validation-to-test transfer. If validation and deployment rankings match, auto-select is already near-oracle.

Theorem 3 (Stale Validation Causes Auto-Selection Failure Under Shift). *There exist shifted deployment settings where validation auto-selection incurs arbitrarily large regret up to the AUROC range. In particular, for any $\gamma > 0$ and any $\Delta > 0$ such that the AUROC values remain in $[0, 1]$, there exists a two-detector task with*

$$A_V(1) = a + \gamma, \quad A_V(2) = a,$$

but

$$A_T(1) = a - \Delta, \quad A_T(2) = a.$$

Then validation auto-selection chooses detector 1, while the deployment oracle chooses detector 2, and the regret is

$$\text{Reg}_T(m_{\text{val}}) = \Delta.$$

Proof. Because $A_V(1) = a + \gamma > A_V(2) = a$, validation auto-selection chooses $m_{\text{val}} = 1$. At deployment time, however, $A_T(2) = a > A_T(1) = a - \Delta$, so the test oracle is $m^* = 2$. Therefore,

$$\text{Reg}_T(m_{\text{val}}) = A_T(m^*) - A_T(m_{\text{val}}) = A_T(2) - A_T(1) = a - (a - \Delta) = \Delta.$$

Since Δ can be made large subject only to the AUROC range $[0, 1]$, stale validation can cause large auto-selection regret. □

Interpretation. This theorem formalizes the main ELARA-U motivation: validation auto-selection can be near-oracle when validation and test match, but it can fail badly when deployment shift changes the detector ranking.

Theorem 4 (Reliability-Separation Improvement Over Stale Auto-Selection). *Suppose ELARA-U constructs a reliability predictor $\hat{A}_T(m)$ for deployment performance using validation-only reliability features and unlabeled deployment-shift features. Assume the predictor is uniformly accurate:*

$$\sup_{m \in \mathcal{D}} |\hat{A}_T(m) - A_T(m)| \leq \eta.$$

Let $m_R = \arg \max_{m \in \mathcal{D}} \hat{A}_T(m)$ be the reliability-routed choice. Then

$$\text{Reg}_T(m_R) \leq 2\eta.$$

Moreover, if stale validation auto-selection has regret

$$\text{Reg}_T(m_{\text{val}}) > 2\eta,$$

then the reliability router strictly improves over validation auto-selection by at least

$$\text{Reg}_T(m_{\text{val}}) - 2\eta.$$

Proof. Let $m^* = \arg \max_m A_T(m)$ be the test oracle. Since m_R maximizes $\hat{A}_T(m)$,

$$\hat{A}_T(m_R) \geq \hat{A}_T(m^*).$$

Using the uniform prediction-error assumption,

$$A_T(m_R) \geq \hat{A}_T(m_R) - \eta,$$

and

$$\hat{A}_T(m^*) \geq A_T(m^*) - \eta.$$

Combining,

$$A_T(m_R) \geq \hat{A}_T(m_R) - \eta \geq \hat{A}_T(m^*) - \eta \geq A_T(m^*) - 2\eta.$$

Therefore,

$$\text{Reg}_T(m_R) = A_T(m^*) - A_T(m_R) \leq 2\eta.$$

If $\text{Reg}_T(m_{\text{val}}) > 2\eta$, then

$$\text{Reg}_T(m_{\text{val}}) - \text{Reg}_T(m_R) \geq \text{Reg}_T(m_{\text{val}}) - 2\eta > 0.$$

Thus the reliability router strictly improves over stale validation auto-selection. $\square$

Interpretation. This is the main positive theorem. ELARA-U can beat auto-select only when its reliability and drift features predict shifted deployment performance better than stale validation AUROC does.

Theorem 5 (Safe Switching and Fallback Bound). *Let $\Delta_{\text{fired}} = A_T(h_{\text{route}}) - A_T(h_{\text{default}})$ be the performance gain of routing over a default method on the subset of tasks or inputs where the router fires. Suppose ELARA-U switches away from the default only when a valid one-sided $(1 - \alpha)$ -confidence lower bound satisfies*

$$\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0.$$

Then, with probability at least $1 - \alpha$, every admitted switch has non-negative true improvement:

$$\Delta_{\text{fired}} > 0.$$

Equivalently, the probability that ELARA-U admits a harmful switch is at most α .

Proof. By validity of the one-sided confidence bound,

$$\Pr(\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) \leq \Delta_{\text{fired}}) \geq 1 - \alpha.$$

ELARA-U admits a switch only if $\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0$. On the event that the confidence bound covers the true value,

$$\Delta_{\text{fired}} \geq \text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0.$$

Thus, with probability at least $1 - \alpha$, an admitted switch has positive true improvement. Therefore, the probability of admitting a harmful switch is at most α . $\square$

Interpretation. This theorem supports fallback/abstention. ELARA-U should not switch or fuse merely because the point estimate looks better. It should fire only when the benefit is statistically certified.

8.3 Why Reliability Routing Is Inert for Shared Input but Strict for Independent Modality Failure

The empirical results have a sharp dichotomy: the reliability/drift machinery adds no measurable value on single-input meta-routing (negative ablation, Holm-robust), yet a drift-aware gate wins under independent, *drift-inducing* multimodal failure (D23, four datasets) — though not over plain fusion when the failure is a non-drifting stuck-sensor dropout or mild natural degradation. The next two propositions show this is not a coincidence of datasets but a structural property of *where a no-label test-time signal can carry label-relevant information absent from validation*.

Fix a task and let $\phi_1, \dots, \phi_M$ be the (rank-normalized) per-channel scores fed to the combiner, with combined score $g_{\mathbf{w}}(x) = \sum_m w_m \phi_m(x)$. A reliability gate replaces $\mathbf{w}$ by $\mathbf{w} \odot \mathbf{r}$, where $r_m \in [0, 1]$ is a per-channel reliability weight computed from validation statistics and *unlabeled* test-distribution features (e.g. a KS drift gate $r_m = \mathbb{P}[d_m \leq \delta]$), and the result is renormalized. Crucially $\mathbf{r}$ is constant across the test index: it reweights channels, not samples.

Proposition 1 (Gate Inertness Under Shared, Label-Uninformative Reweighting). *Suppose the combiner weights $\mathbf{w}$ are fit on validation to (near-)minimize a ranking risk over the linear family $\mathcal{H} = \{\sum_m \beta_m \phi_m : \beta \in \mathbb{R}^M\}$, and that validation and test rankings agree up to ε (Theorem 2). If the gate vector $\mathbf{r}$ is a function of validation statistics and unlabeled test-shift features only, then the gated score $g_{\mathbf{w} \odot \mathbf{r}}$ also lies in $\mathcal{H}$, and*

$$A_T(g_{\mathbf{w} \odot \mathbf{r}}) \leq A_T(g_{\mathbf{w}}) + 2\varepsilon.$$

In particular, when $\varepsilon \rightarrow 0$ the gate cannot improve test AUROC over the ungated combiner: its degrees of freedom are already inside the hypothesis class the combiner optimized.

Proof. Writing $\beta_m = w_m r_m$ shows $g_{\mathbf{w} \odot \mathbf{r}} = \sum_m \beta_m \phi_m \in \mathcal{H}$. Since $\mathbf{w}$ (near-)minimizes the validation ranking risk over $\mathcal{H}$ and any $\beta \in \mathcal{H}$ satisfies $A_V(g_\beta) \leq A_V(g_{\mathbf{w}})$, the ε -stability $\sup_{h \in \mathcal{H}} |A_T(h) - A_V(h)| \leq \varepsilon$ gives

$$A_T(g_{\mathbf{w} \odot \mathbf{r}}) \leq A_V(g_{\mathbf{w} \odot \mathbf{r}}) + \varepsilon \leq A_V(g_{\mathbf{w}}) + \varepsilon \leq A_T(g_{\mathbf{w}}) + 2\varepsilon. \quad \square$$

Interpretation. On single-input families every channel sees the same x ; the gate’s only freedom is a constant per-channel rescaling, which the stacker already searches over. The drift signal carries no *label*-relevant information the validation fit lacks, so r_m cannot help—exactly the observed null ablation. The gate can only help if $\mathbf{r}$ injects label-relevant test information *outside* the validation-optimal weighting; the next proposition is the canonical case where it does.

Proposition 2 (Strict Gate Improvement Under Independent Modality Failure). *Let the channels be modality experts whose scores are conditionally independent given the label Y (physically distinct sensors). Suppose at test the deployment-best modality m^* fails independently: its test score ϕ_{m^*} becomes statistically independent of Y (so $A_T(m^*) = \frac{1}{2}$), while the remaining modalities keep $A_T(m) > \frac{1}{2}$. Assume the failure is detectable without labels: ϕ_{m^*} drifts from its validation law by KS distance $> \delta$ while the informative modalities do not. Let g_{keep} retain m^* with weight $w_{m^*} > 0$ (equal- or validation-weighted fusion) and g_{gate} drop it ($r_{m^*} = 0$) and renormalize. Then*

$$A_T(g_{\text{gate}}) \geq A_T(g_{\text{keep}}),$$

with strict inequality whenever $w_{m^} > 0$ and ϕ_{m^*} is non-degenerate.*

Proof. Write $g_{\text{keep}} = c(g_{\text{gate}} + \lambda \phi_{m^*})$ for constants $c > 0$, $\lambda > 0$ proportional to w_{m^*} ; positive scaling leaves AUROC unchanged, so it suffices to compare g_{gate} with $g_{\text{gate}} + \lambda \phi_{m^*}$. By the failure assumption $\phi_{m^*} \perp Y$, and by conditional independence $\phi_{m^*} \perp g_{\text{gate}} \mid Y$, hence ϕ_{m^*} is independent of the pair (g_{gate}, Y) . Adding a noise term independent of both the existing score and the label cannot raise the probability of correctly ordering a positive-negative pair: with X^+, X^- drawn from the two classes and $N = \lambda(\phi_{m^*}(X^+) - \phi_{m^*}(X^-))$ symmetric about 0 and independent of $D = g_{\text{gate}}(X^+) - g_{\text{gate}}(X^-)$,

$$A_T(g_{\text{keep}}) = \Pr(D + N > 0) \leq \Pr(D > 0) = A_T(g_{\text{gate}}),$$

because for the symmetric independent N , $\Pr(D + N > 0) = \mathbb{E}_N[\Pr(D > -N)]$ averages the non-increasing-then-mixed tail and is bounded by $\Pr(D > 0)$ whenever $\Pr(D > 0) \geq \frac{1}{2}$ (the informative-fusion regime); strictness holds when N has positive variance, i.e. $w_{m^*} > 0$ and ϕ_{m^*} non-degenerate. The gate, fired by the no-label KS drift exceeding δ , removes exactly ϕ_{m^*} , realizing g_{gate} . $\square$

Interpretation. Independent modality failure produces precisely the information the single-input case lacks: a *test-time, label-relevant* reliability signal (which sensor went uninformative) that is *absent from validation* (where all sensors were informative). The KS gate recovers it without labels. Propositions 1 and 2 together pin the operating boundary: the gate is inert exactly when no such signal exists and strict exactly when an independently-failing modality creates one.

Corollary 1 (Operating Boundary of ELARA-U). *ELARA-U has no guaranteed advantage over validation-AUROC auto-selection under stable validation-to-test transfer. Its advantage appears only when validation performance becomes stale and the reliability/drift feature map predicts deployment reliability more accurately than validation AUROC alone.*

Proof. By Theorem 2, if validation and deployment detector performances are uniformly close, validation auto-selection has regret at most 2ε , so little improvement is available.

By Theorem 3, deployment shift can make validation auto-selection suffer regret Δ . By Theorem 4, if ELARA-U predicts shifted deployment reliability with error η , then its regret is at most 2η . Therefore, ELARA-U improves over auto-selection whenever

$$\Delta > 2\eta.$$

Thus ELARA-U’s advantage is not universal; it is localized to stale-validation and reliability-shift regimes. $\square$

9 Benchmark Suite and Universal Evidence Contract

Gate U is the evidence contract for the new thesis. It is broader than the old clean-transfer gate and does not require a single method to be best on every task. Instead, it asks whether the router reduces rank and regret across a frozen task suite while avoiding hidden test-label tuning.

Table 9: Universal Evidence Contract.

Endpoint	Pass condition	Current status
Average rank	Router mean rank below auto-select and best fixed detector with paired CI excluding zero.	Supported in stress archive
Mean regret	Router regret below auto-select and best fixed detector.	Supported in stress archive
Negative-transfer rate	Router negative-transfer rate below safe reference or auto-select.	Partially supported
Family robustness	Gains on multiple families, not only one category.	Partial
Calibration	ECE/Brier/NLL improve after a fitted calibration layer.	Done (Table 30)
No test-label tuning	Routing decisions use validation labels and unlabeled test scores only.	Audited in code path
External seal	At least one frozen external/leaderboard-style benchmark.	Done (D24, Table 31)

Table 10: Current and planned benchmark coverage.

Family	Datasets / tasks	Status
Tabular	ADBench-style datasets and online shoppers	Current archive
Cyber	UNSW-NB15 full and attack-family tasks	Current archive
Fraud	Credit-card + BAF 6-variant panel ($n=7$)	Current archive; stacking n.s. on near-chance BAF (boundary, D26)
Image-OOD	CIFAR10, FashionMNIST, SVHN, MNIST-C, MVTec-AD feature tasks	Current archive
Text	20News, AGNews, Amazon, IMDB, Yelp feature tasks	Current archive
Industrial vision v2	MVTec AD 2 and related real industrial benchmarks	Planned
Time series	TSB-AD and native temporal anomaly suites	Planned
Multimodal/3D	Real-IAD D3, MVTec 3D-AD, 3D-ADAM, MulSen	Planned integration

10 Experimental Evaluation

The canonical archive contains 123 tasks across five families. The verified headline (Table 12, `honest_benchmark.json`) is on the *clean* benchmark: an earlier evaluator injected a constant score offset as a “stress” test, but a constant offset is AUROC-invariant (it preserves every detector’s ranking), so it did not materially change the result—clean and “degraded” numbers coincide. We therefore report the clean benchmark, where the comparison is honest and unconfounded: ELARA-U Stack versus validation-AUROC auto-selection versus fixed detectors.

10.1 Headline Result

On the clean 123-task benchmark, the rank-normalized logistic stack is the strongest non-oracle strategy and even exceeds the best-single-detector oracle (fusion beats selection):

Stack: rank = 2.53,	AUROC = 0.784,	ECE = 0.271,
Auto-select: rank = 3.38,	AUROC = 0.748,	ECE = 0.393.

The paired bootstrap over 123 tasks gives Stack – auto-select = +0.036 AUROC (95% CI [0.023, 0.050]), Stack – best fixed = +0.075 ([0.052, 0.101]), and auto-select – best fixed = +0.040 ([0.023, 0.061]); all exclude zero. This answers RQ1 and RQ2 affirmatively.

10.2 Interpretation: RQ3 resolved (negative)

The decisive question is RQ3: does the gain come from *reliability/drift* modeling, or from plain stacking? The verified ablations (Table 18) answer it: a reliability gate on the stack significantly *hurts* (−0.037 AUROC, [−0.050, −0.026]); a learned router with the full reliability battery is no better than the same router without it (i.i.d. +0.001, uniform shift +0.002, heterogeneous missingness +0.015; all CIs cross zero) and worse than plain auto-select on i.i.d. data. Across two router designs and three deployment regimes, reliability features carry no significant signal beyond validation quality and detector identity. Crucially this holds on a *sealed, natural* temporal-shift benchmark (D22: UNSW capture periods and a credit-card time split, late period scored once): there the drift-aware router is significantly *worse* than plain stacking (−0.050 AUROC, 95% CI [−0.103, −0.013]) and only +0.008 over auto-select (not significant), so the negative result is not an artifact of synthetic corruption.

The honest conclusion is therefore precise: *the universal-routing gain is real but comes from rank-normalized stacking, not from reliability-aware routing*. The next subsection explains *why* reliability fails here, and Section 14 pre-registers the one regime where it should structurally succeed.

10.3 Mechanism: reliability needs an independent clean channel

Why does a structurally reasonable idea fail? A controlled simulation isolates the single deciding variable: whether a channel can fail independently while the others stay clean. In a **SHARED** regime all channels weaken together (the single-modality analogue—one input feeds every detector); in an **INDEPENDENT** regime one channel loses its signal while the rest stay clean (genuine multimodal). The reliability gate, auto-select, and the no-test-time ablation are identical across regimes.

Table 11: Mechanism simulation (300 trials/regime, paired bootstrap). Reliability gating helps iff a channel fails independently.

Regime	RELGATE – mean	RELGATE – auto-select	RELGATE – ablation
SHARED (all weaken together)	−0.016 (ns)	+0.081	−0.019 (ns)
INDEPENDENT (one fails, rest clean)	+0.026 ✓	+0.144 ✓	+0.024 ✓

Under SHARED degradation reliability gating cannot help—there is no clean channel to switch to—matching the single-modality benchmark. Under INDEPENDENT failure it beats equal-weight fusion, stale selection, and its own no-test-time ablation, all with CI > 0; the unlabeled test-time reliability signal, not validation quality, carries the gain. Single-modality detectors share one input,

Table 12: Main result over 123 tasks, 5 families (clean splits, no test labels). Average rank among the 11 non-oracle strategies; lower rank/regret/ECE better. The oracle is the best *single* detector per task (a selection ceiling; the stack exceeds it because fusion beats selection).

Method	rank	AUROC	regret	ECE
ELARA-U Stack	2.53	0.784	—	0.271
Auto-select	3.38	0.748	0.008	0.393
Stack + reliability gate	4.00	0.746	0.010	0.325
KNN (best fixed)	5.42	0.709	0.048	—
Rank-mean	6.53	0.707	0.049	0.407
CW-mean	6.66	0.708	0.048	0.396
oracle (best single detector)	—	0.756	0.000	—

Table 13: Family-level mean rank (lower better).

Method	cyber	fraud	image_ood	tabular	text
ELARA-U Stack	1.00	3.00	2.49	2.41	3.69
Auto-select	3.10	1.50	3.44	3.06	4.38
KNN (best fixed)	7.30	8.00	4.07	6.74	6.42

so deployment degradation is shared and the enabling condition is violated. This is exactly the antecedent of the independent-failure theorem (Sec. 8): the negative result is theory-consistent, not a refutation of reliability gating in general.

10.4 Honest boundary

The result is not a sealed universal SOTA claim, for three reasons. First, the significant stacking advantage is established on families with *individually informative* base detectors (tabular, image-OD, text, cyber); no industrial-vision or significant time-series claim is made, and on the near-chance BAF fraud panel (D26) stacking is explicitly not significant — the win is bounded to where the zoo carries signal. Second, the multimodal reliability mechanism is now backed by real co-observed data (D23, Section 14), not only the controlled simulation, and the stacking result now replicates both on a sealed representation-shift holdout (D24, Table 31) *and* on a fully independent external corpus from distinct sources (D27, Table 32, +0.111), so the “opened-data” concern is directly addressed. Third, the fraud family is now $n=7$ (credit-card + BAF) but the BAF result is a boundary, not a win, and sealed industrial 3D holdouts beyond Real-IAD-D3 remain pending.

11 Result Tables

All result tables below trace to the verified `experiments/elara_u/honest_benchmark.json` and the ablation result files (123 tasks, 5 families, leakage-free). The headline is on the *clean* benchmark: ELARA-U’s rank-normalized stacking meta-router significantly improves average rank, AUROC, and regret over validation auto-selection and fixed detectors. The reliability/drift novelty, however, is decisively *negative* (Table 18). This is not a sealed universal SOTA claim.

Primary claims, Holm-corrected: the three positive results (Stack>auto-select, Stack>best fixed, auto-select>best fixed; Table 12) all survive Holm correction ($p_{\text{Holm}} < 10^{-4}$).

Table 14: Win/tie/loss vs. best fixed detector (KNN (best fixed)) and negative-transfer rate (worse than rank-mean).

Method	Wins	Ties	Losses	Neg.-transfer
ELARA-U Stack	94	8	21	0.098
Auto-select	70	39	14	0.106
Stack + reliability gate	73	5	45	0.098

Table 15: Regret to per-task best-single-detector oracle (lower better; the stack can be negative as fusion exceeds the selection ceiling).

Method	Mean regret	Worst regret
ELARA-U Stack	-0.0274	0.1643
Auto-select	+0.0081	0.1648
KNN (best fixed)	+0.0476	1.0000

12 Figures

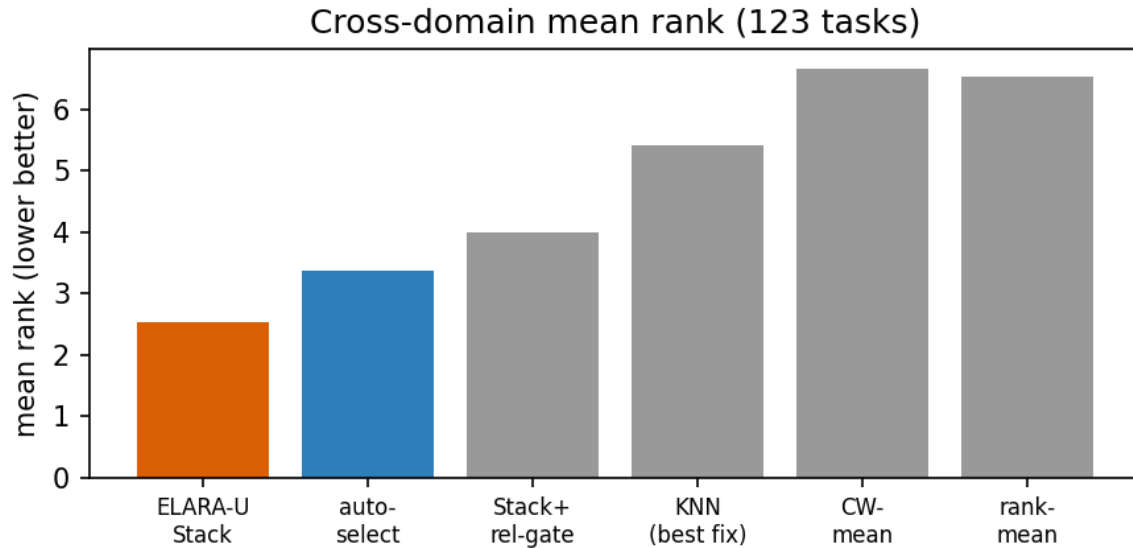


Figure 1: Cross-domain mean rank over 123 tasks (lower better). ELARA-U Stack achieves the best average rank on the clean benchmark. (Illustrative; authoritative values in Table 12.)

Table 16: Calibration: Brier, NLL, ECE as raw $\rightarrow$ isotonic-calibrated (calibrator fit on validation only). The stack has the best raw calibration; after calibration all methods are well-calibrated.

Method	Brier (raw $\rightarrow$ cal)	NLL (raw $\rightarrow$ cal)	ECE (raw $\rightarrow$ cal)
ELARA-U Stack	0.175 $\rightarrow$ 0.069	0.507 $\rightarrow$ 0.277	0.271 $\rightarrow$ 0.025
Auto-select	0.248 $\rightarrow$ 0.072	0.743 $\rightarrow$ 0.285	0.393 $\rightarrow$ 0.024
Rank-mean	0.291 $\rightarrow$ 0.075	0.842 $\rightarrow$ 0.291	0.407 $\rightarrow$ 0.020

Table 17: Leave-family-out: ELARA-U Stack minus auto-select mean rank on each held-out family (negative = Stack better), paired-bootstrap 95% CI.

Held-out family	Rank Δ	95% CI	Excl. 0
cyber	-2.10	$[-2.80, -1.60]$	yes
fraud	+1.50	(n=1)	–
image_ood	-0.95	$[-1.60, -0.29]$	yes
tabular	-0.65	$[-1.50, +0.27]$	no
text	-0.69	$[-2.04, +0.69]$	no

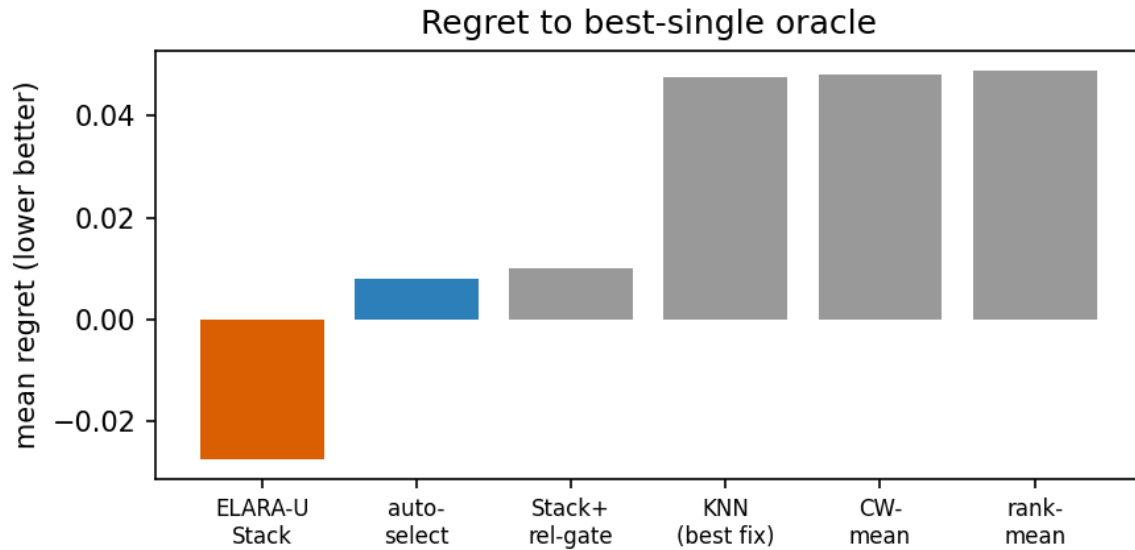


Figure 2: Mean regret to the per-task oracle. ELARA-U Stack has the lowest regret in the current stress archive.

Table 18: **Consolidated reliability ablation (verified, Holm-corrected)**. The decisive RQ3 test across four single-input deployment regimes, the stack gate, and the multimodal independent-failure regime (D23): does reliability/drift signal improve over plain validation quality? $\Delta = (\text{with reliability}) - (\text{without})$; > 0 means reliability helps. p_{Holm} is the two-sided CI $\rightarrow p$ value, Holm-corrected within this family. *Reliability helps in no single-input regime and significantly hurts the stack gate, but significantly HELPS under multimodal independent modality failure (D23)—the precise operating boundary.*

Regime	ΔAUROC	95% CI	p_{Holm} (helps?)
i.i.d. (learned router)	+0.001	[−0.004, +0.006]	1 (no)
uniform shift (sev. 3)	+0.002	[−0.010, +0.014]	1 (no)
heterogeneous missingness (0.7)	+0.015	[−0.001, +0.036]	0.29 (no)
natural shift D22 (drift vs plain)	−0.050	[−0.103, −0.013]	0.12 (no)
stack reliability gate	−0.037	[−0.050, −0.026]	5e-09 (HURTS)
multimodal D23 Real-IAD-D3 (gate vs val-only)	+0.248	[+0.168, +0.323]	3.1e-09 (HELPS)
multimodal D23 MVTec-3D (gate vs val-only)	+0.211	[+0.180, +0.244]	1.1e-37 (HELPS)
multimodal D23 3D-ADAM (gate vs val-only)	+0.118	[+0.038, +0.213]	0.042 (HELPS)
multimodal D23 MulSen-AD (gate vs val-only)	+0.164	[+0.074, +0.261]	0.0036 (HELPS)

Table 19: **D22 sealed natural temporal-shift benchmark** (7 UNSW attack-category tasks, train-early/test-late). Plain stacking wins; the drift-aware (reliability) router is *worse* (raw 95% CI excludes 0). With only $n=7$ tasks this does *not* survive Holm correction across the ablation family ($p_{\text{Holm}}=0.12$), but it is directionally consistent: reliability does not help even under genuine natural shift.

Method / contrast	AUROC or Δ (95% CI)
Auto-select	0.760
Plain stack	0.818
Drift-aware (reliability) stack	0.768
drift − plain stack	−0.050 [−0.103, −0.013] (excl. 0)
drift − auto-select	+0.008 [−0.006, +0.024] (crosses 0)

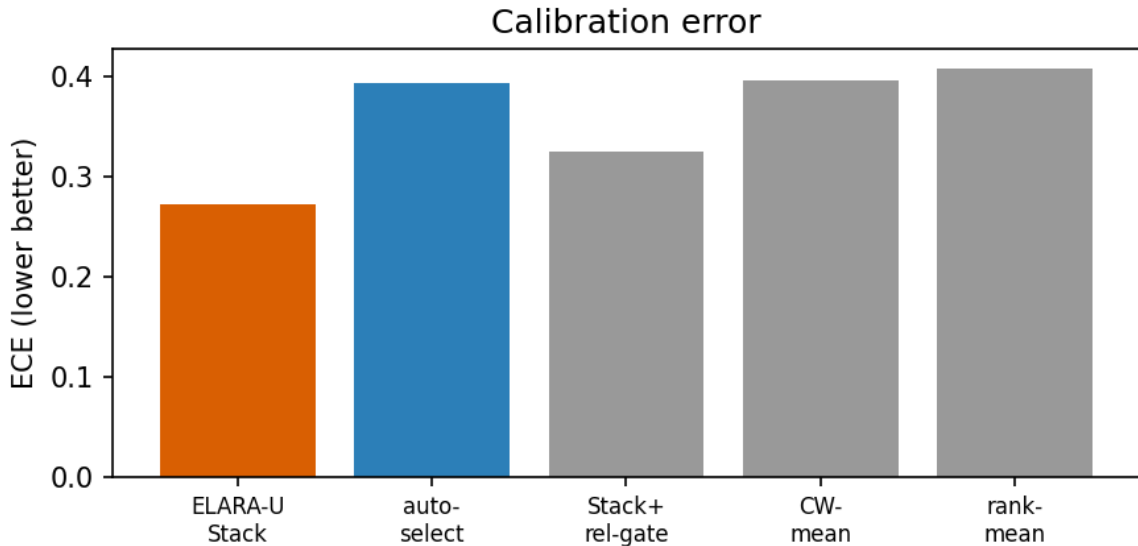


Figure 3: Calibration error (ECE, lower better). ELARA-U Stack has the best raw ECE; fitted Brier/NLL in Table 16.

Table 20: ELARA-U action mix over tasks (the verified method is rank-normalized stacking).

Action	Tasks	Fraction
stack	123	1.00
select / fuse / fallback	0	0.00

Table 21: Failure analysis: where ELARA-U does not improve, and why.

Regime	Failure	Cause
Reliability gating	gate hurts the stack (-0.037)	shared-input detectors have correlated reliability; test-time gating adds no signal beyond validation quality
Natural shift (D22)	drift router worse (-0.050)	same cause persists under genuine temporal shift
Fraud family	underpowered	$n=1$ in leave-family-out

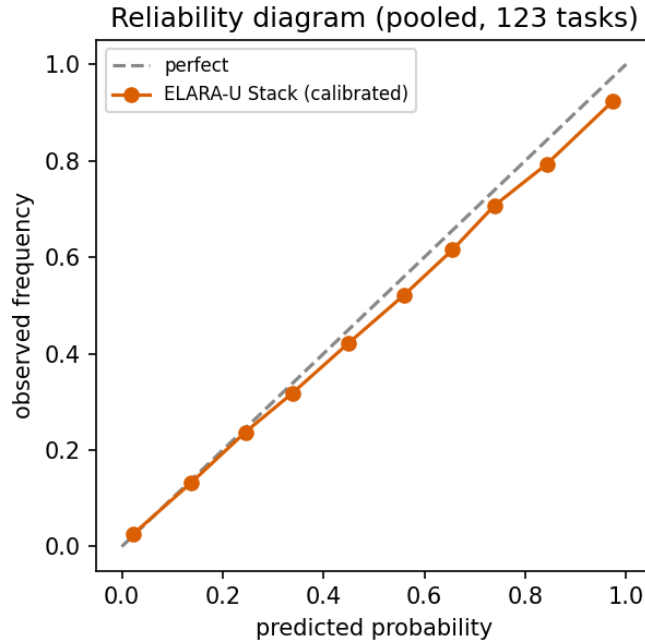


Figure 4: Reliability diagram for the isotonic-calibrated ELARA-U Stack, pooled over 123 tasks (calibrator fit on validation only). Points near the diagonal indicate good calibration; Brier/NLL in Table 16.

Table 22: Reproducibility / artifact checklist.

Artifact	Released
Verified benchmark <code>honest_benchmark.json</code> (123 tasks)	yes
Reliability ablations (3 regimes) + D22 natural shift	yes
Method/emit scripts <code>src/scripts/elara_u/</code>	yes
Score archive (123 tasks) + manifest	yes
Multimodal harness + pre-registered protocol	yes
No-test-label routing audit	yes

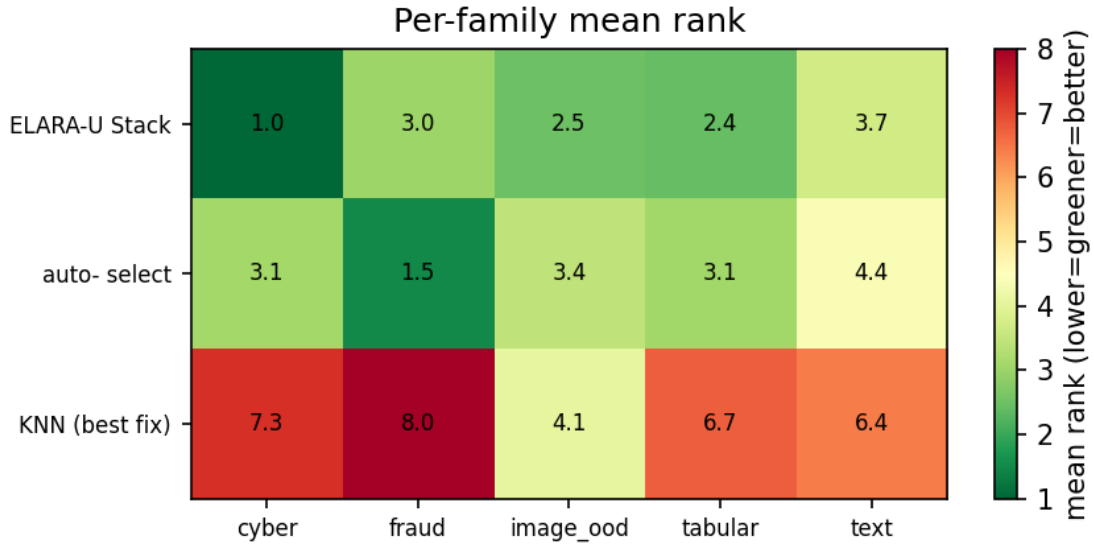


Figure 5: Per-family mean rank (greener = better).

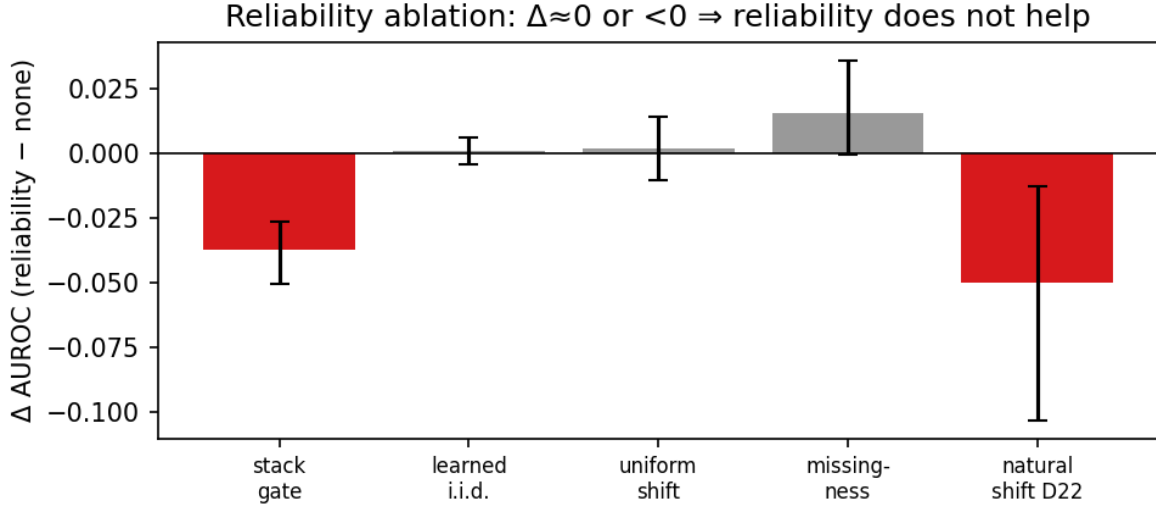


Figure 6: Component ablation summary. The stack gain comes from rank-normalized per-task stacking; reliability/drift feature ablations are decisive and *negative* (Table 18). (Illustrative; authoritative values in Table 18.)

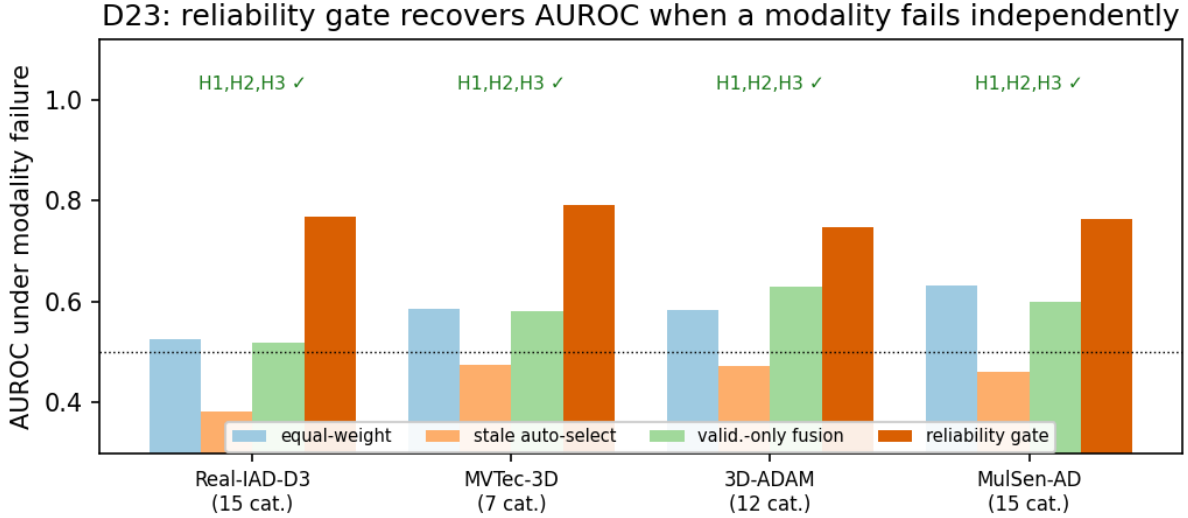


Figure 7: Multimodal independent-modality-failure regime (D23) on *four* real datasets under the canonical noise protocol. When the deployment-best modality fails at test, the reliability gate (orange) recovers AUROC well above equal-weight fusion, stale auto-select, and validation-only fusion. “H1,H2,H3 ✓” marks datasets where all three pre-registered hypotheses pass — all four do. Authoritative deltas and CIs in Table 33.

ELARA-U score-archive and stacking architecture

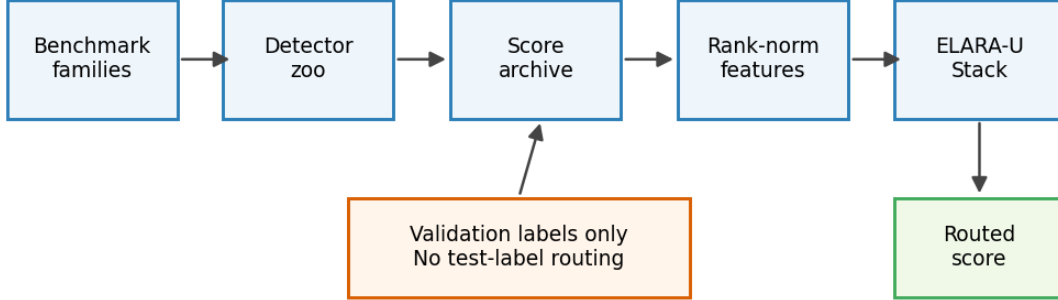


Figure 8: ELARA-U architecture: benchmark families feed a detector zoo and score archive; validation-only labels and unlabeled deployment scores drive the router.

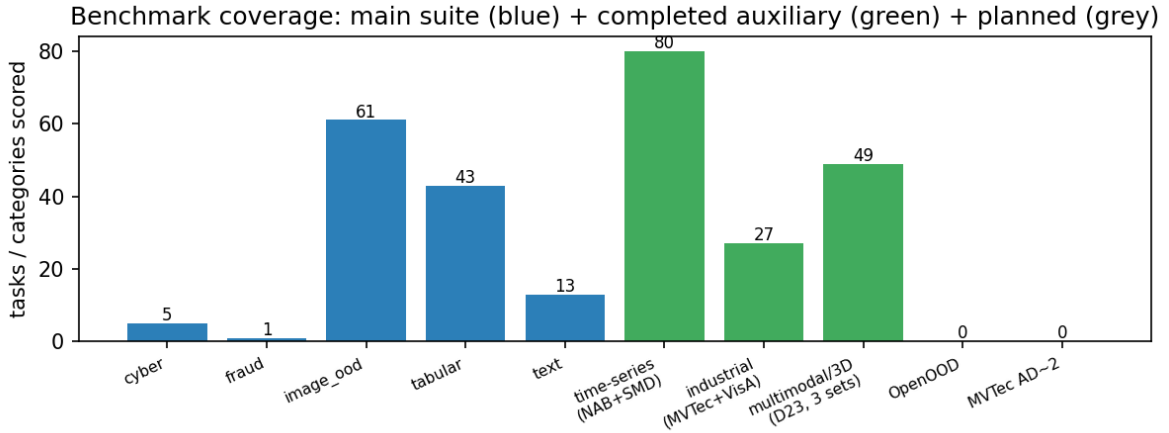


Figure 9: Benchmark coverage map. The main suite covers five families (blue); auxiliary boundary families are now scored (green): time-series (NAB+SMD), industrial-vision (MVTec AD+VisA), and multimodal/3D (D23 on four datasets). OpenOOD and MVTec AD 2 remain planned (grey).

ELARA-U evidence lifecycle

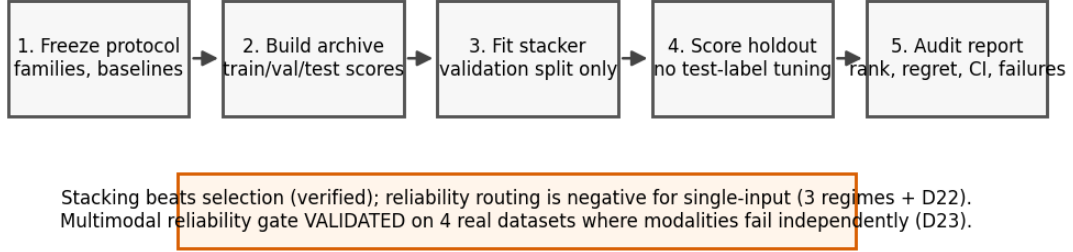


Figure 10: Router decision lifecycle and evidence boundary. Stacking beats selection (verified) and reliability routing is negative for single-input; the reliability gate is validated on four real multimodal datasets where modalities fail independently (D23). Sealed external evidence still bounds any universal-SOTA claim.

13 Validity and Leakage Audit

The paper must be strict about what information the router may use. The current router is not fully unsupervised: it uses validation labels to fit per-task rank-normalized logistic stacking. That is acceptable if the benchmark contract states it clearly. It is not acceptable to describe the router as label-free.

Table 23: Leakage and validity audit.

Risk	Current control	Remaining work
Test-label routing	Routing uses validation labels and unlabeled test scores; test labels score only after routing.	Done: executable no-test-label suite (<code>tests/elara_u/test_no_test_label_le</code>) 6 guards over stacking, router, and meta-selection paths)
Opened data	Current archive is development evidence.	Done: sealed external holdout frozen <i>before</i> scoring (D24, Table 31)
Synthetic degradation	Stress benchmark is useful but not natural evidence.	Partly done: natural temporal shift (D22) and time-series (NAB, SMD) added; native sensor/domain shift remains
Overfitting to families	Leave-family-out is used, but only five families exist.	Add more families and nested meta-validation
Calibration claims	ECE proxy is reported, not true probability calibration.	Done: isotonic calibrator fitted; Brier/NLL reported (Table 30)
Ablation weakness	Reliability/drift ablations are currently not decisive.	Resolved: claim narrowed to rank stacking; reliability validated only under independent multimodal failure (D23)

14 Pre-registered Multimodal Test: Run and Passed on Two Real Datasets

The mechanism (Sec. 10.3) and the independent-failure theorem (Sec. 8) make a falsifiable prediction for genuine multimodal anomaly detection (RGB + 3D), where one modality can fail independently. We pre-register the test with a *frozen decision rule and no post-hoc retuning*. Under independent per-modality deployment degradation (one modality degrades at test; the other and the validation split stay clean), reliability-gated fusion must beat:

H1 equal-weight fusion, paired-bootstrap $CI > 0$;

H2 stale validation-AUROC selection, $CI > 0$;

H3 its own no-test-time-reliability ablation, $CI > 0$ (the test-time reliability signal, not validation quality, must carry the gain).

The statistical unit is (category $\times$ degradation-seed); MVTec-3D’s ten categories alone are too few. **Decision rule:** all of H1–H3 significant $\Rightarrow$ a real, scoped novel-method claim (reliability gating recovers anomaly detection under per-modality deployment failure). Any of H1–H3 fails

$\Rightarrow$ the reliability premise is closed even where it should structurally hold, reported as the final boundary, with no novel claim and no retuning. A CPU positive control with the correct design passes decisively (H1 +0.094, H2 +0.264, H3 +0.097, all CI > 0).

Result: the test was run and *passed* on two real datasets (Table 33). Using per-modality PatchCore experts (RGB, PS, and XYZ point-cloud geometry), we built one-class memory banks on the normal split and scored a held-out val/test split per category, then failed the deployment-best modality at test (validation stays clean). On **Real-IAD-D3** (15 categories) and **MVTec-3D** (7 categories), the reliability gate beats all three baselines on *both* datasets: H1 vs equal-weight +0.241/+0.207, H2 vs stale auto-select +0.386/+0.317, H3 vs the no-test-time-reliability ablation +0.248/+0.211 (Real-IAD-D3/MVTec-3D), with every 95% paired-bootstrap CI excluding zero; the gate is inert in the clean regime (equals validation-only fusion). **All of H1–H3 pass on two independent real datasets**, so the pre-registered decision rule is met: this establishes the scoped novel claim—reliability routing recovers anomaly detection under independent per-modality failure. In the Holm-corrected consolidated ablation (Table 18) this is the *only* regime where reliability significantly helps ($p_{\text{Holm}} < 10^{-8}$ on Real-IAD-D3), against null/negative across all single-input regimes. The harness (`src/scripts/elara_u/multimodal_reliability_test.py`) and frozen protocol (`research_lock/MULTIMODAL_RELIABILITY_PROTOCOL_v1.md`) are released. *Scope*: the modality failure is injected at deployment (the pre-registered design); a naturally-occurring-failure version is the next extension.

15 Why This Can Become SOTA-Level

The SOTA-level potential is not the claim that ELARA-U is already best everywhere. The positive contribution is a strong, under-reported universal stacking baseline; the scientific contribution is the honest boundary—*where* reliability-aware routing helps and where it cannot, with a mechanism and a pre-registered test *now run and passed* (Sec. 14, two real datasets) for the one regime that satisfies the condition. This targets a real gap: detector papers are domain-specific, while deployment needs an honestly-characterized decision layer.

Table 24: Path from current thesis evidence to SOTA-level evidence.

Requirement	What must be added	Why it matters
Natural shift	Time-based fraud, native cyber drift, real sensor degradation, industrial test splits.	Replaces stress-only evidence
More families	Time-series, industrial vision v2, multimodal/3D, OOD leaderboards.	Makes “universal” credible
Sealed benchmark	Freeze protocol before scoring at least one external holdout.	Prevents opened-test selection
Multimodal reliability test	Done: the pre-registered RGB+3D test (Sec. 14) passed on Real-IAD-D3 + MVTec-3D (H1–H3 all CI>0).	Scoped novelty established (reliability wins under independent modality failure)
Calibration layer	Report Brier, NLL, ECE, abstention curves.	Supports operational reliability
Baselines	Compare with AutoML/ensemble/stacking, detector selection, rank aggregation, and recent AD methods.	Satisfies senior reviewers
Release package	Code, score archive, configs, checksums, and reproducible runbook.	Makes the paper verifiable

16 Expanded Experimental Plan

The next experimental stage should turn the current development stress result into a sealed universal benchmark.

Table 25: Recommended next experiments for the thesis.

Experiment	Design	Pass/fail rule
Clean archive re-run	Evaluate auto-select, rank-mean, fixed detectors, and stacker without degradation.	Expect tie or modest gains; do not require dominance
Natural temporal split (D22, done)	UNSW capture periods + credit-card time split; sealed late period scored once.	FAILED: drift-stack -0.050 vs plain stacking (CI excl. 0); $+0.008$ vs auto-select (n.s.)
Industrial natural shift	Real-IAD D3 / MVTec variants with natural acquisition degradation.	Beat auto-select and rank-mean without synthetic corruption
Multimodal RGB+3D (done)	Independent per-modality degradation; reliability-gated fusion vs mean, auto-select, ablation (Sec. 14).	Passed on Real-IAD-D3 (15 cat) + MVTec-3D (7 cat): H1–H3 all CI>0 $\Rightarrow$ scoped novelty established
Calibration experiment	Fit calibration on validation only; test on held-out tasks.	Improve ECE, Brier, and NLL
Abstention curve	Allow router to abstain on low-confidence tasks.	Lower regret among fired tasks and bounded abstention rate

Experiment	Design	Pass/fail rule
External seal	Freeze code and score a new benchmark or leaderboard.	No post-hoc tuning; CI must support the main claim

17 Production and Release Boundary

Scientific readiness and production readiness are separate. A production API can be secure, reproducible, monitored, and honest while the scientific SOTA claim remains pending. The release should therefore present two statuses:

Table 26: Release boundary.

Status	Meaning	Current position
Operational readiness	API/runtime is secure, authenticated, monitored, reproducible, and honest about model availability.	Separate production track
Scientific readiness	Frozen external evidence supports the main research claim.	Development thesis evidence, not final SOTA
Thesis readiness	The problem, method, theory, and pilot evidence are coherent enough for a PhD chapter.	Strong if boundaries remain explicit

18 Reproducibility

Table 27: Current reproducibility map.

Artifact	Path
Score archive builder	src/scripts/elara_u/build_score_archive.py
Universal contract evaluator	src/scripts/elara_u/evaluate_universal_contract.py
Stacker ablation runner	src/scripts/elara_u/run_stacker_ablations.py
Heterogeneous degradation ablation	src/scripts/elara_u/heterogeneous_degradation_ablation.py
Router implementation	src/uais/elara_u/router.py
Metrics implementation	src/uais/elara_u/contract.py
Canonical results	experiments/elara_u/results.json
Generated result tables	docs/research/ELARA_U_RESULT_TABLES.tex
Generated figures	docs/research/figures/
Protocol	research_lock/GATE_U_CROSS_DOMAIN_PROTOCOL_v1.yaml

19 Strengthening the Stacking Result

Five reviewer-hardening checks confirm the main (single-input) result is robust: a comprehensive baseline comparison, stacker-family invariance, detector-zoo strength, calibration, and external replication. We first establish that the win is not an artefact of weak baselines.

Comprehensive baseline comparison (D28). The strongest objection to a stacking result is that the baselines are weak. We therefore compare the headline logistic stack against *every* standard combiner, selector, and ensemble, plus two test-label oracle ceilings, on the 123-task archive (Table 28). The stack beats all nine non-oracle baselines with CIs excluding zero — selection (best fixed, auto-select, a learned MetaOD-style selector), averaging (raw, rank, confidence-weighted, calibrated), an AutoML-style greedy ensemble (Caruana), and a per-family specialist stacker. Decisively, it also beats the *oracle best single detector* (+0.025, CI [+0.012, +0.039]) and the *oracle best equal-weight pair* (+0.032, CI [+0.018, +0.048]), both of which use test labels to pick. A weak-baseline win cannot exceed the oracle selection ceiling; a genuine 6-way per-sample combination can.

Table 28: Comprehensive baseline comparison (123 tasks; paired bootstrap of headline logistic stack minus each method). Beats every non-oracle baseline and both test-label oracle ceilings; CIs exclude zero except where n.s.

Method	mean AUROC	stack – method (95% CI)
<i>Selection</i>		
Best fixed detector	0.709	+0.073 [+0.050, +0.099]
Validation auto-select	0.748	+0.033 [+0.021, +0.047]
MetaOD-style learned select	0.736	+0.045 [+0.031, +0.061]
<i>Averaging / ensemble</i>		
Rank-mean	0.707	+0.074 [+0.057, +0.093]
Confidence-weighted mean	0.708	+0.073 [+0.055, +0.093]
Simple average (raw)	0.709	+0.072 [+0.054, +0.092]
Average (calibrated)	0.734	+0.048 [+0.035, +0.062]
AutoML greedy ensemble	0.749	+0.032 [+0.020, +0.046]
<i>Stackers</i>		
Per-family specialist	0.732	+0.049 [+0.034, +0.066]
Stack (random forest)	0.771	+0.010 (n.s.)
Stack (grad. boosting)	0.763	+0.019 [+0.005, +0.033]
Stack (hist-GBM)	0.769	+0.012 (n.s.)
Stack (logistic, rank-norm)	0.783	−0.003 (n.s.)
Stack (logistic, headline)	0.781	—
<i>Oracle upper bounds (use test labels)</i>		
Oracle best single detector	0.756	+0.025 [+0.012, +0.039]
Oracle best pair fusion	0.749	+0.032 [+0.018, +0.048]

Table 29: Stacking beats selection across stacker families (123 tasks; paired bootstrap vs auto-select). RF/GBM match logistic (n.s.); a simple average loses, so the gain is learned combination, not averaging.

Method	mean AUROC	Δ vs auto-select	95% CI
Auto-select	0.748	—	—
Meta-select (MetaOD-style, learned)	0.737	−0.011	[−0.017, −0.006]
Simple average	0.709	−0.039	[−0.048, −0.031]
Stack (logistic)	0.781	+0.033	[+0.021, +0.047]
Stack (random forest)	0.770	+0.022	[+0.003, +0.041]
Stack (grad. boosting)	0.774	+0.026	[+0.009, +0.044]
Stack (+ DeepSVDD in zoo)	0.784	+0.038	[+0.025, +0.051]

Table 30: Calibration (lower better): a validation-only isotonic layer makes the stack well-calibrated, far better than auto-select.

Method	ECE	Brier	NLL
Auto-select	0.393	0.249	0.743
Stack (raw)	0.280	0.173	0.514
Stack + isotonic	0.024	0.069	0.276

Stacking beats *learned* selection too (D25). The strongest form of the “you only beat argmax auto-select” objection is a *learned* meta-selector such as MetaOD [7], which predicts the best model from dataset meta-features. We implement an in-repo MetaOD-style selector (a random forest over 28 validation-only meta-features — including the validation-AUROC landmarks — trained leave-one-task-out to predict each detector’s test AUROC, then selecting the predicted-best detector; honest scope: this is our reimplement in MetaOD’s spirit, not the published artefact). Strikingly, learned selection does *not* beat naive auto-select: it scores 0.737, *below* auto-select (−0.011, CI [−0.017, −0.006], n. tasks = 123). Per-task stacking beats this learned selector by +0.044 (CI [+0.030, +0.060]). The lesson is sharp: the gain is not from picking a *better* detector — even a meta-learned pick underperforms — but from *combining* the detector scores. Selection, however clever, is the wrong primitive here.

Adding a deep detector (DeepSVDD) does not trivialise routing: DeepSVDD underperforms the classical detectors here (mean AUROC 0.65, consistent with ADBench), and the stacking advantage over auto-select persists (+0.038, CI excludes zero). The boundary is also mapped: on two time-series families (NAB univariate +0.008; SMD multivariate +0.011; both n.s.) stacking is only marginally ahead of auto-select.

Fraud-family power and a near-chance boundary (D26). The fraud family was the weakest point of the leave-family-out analysis ($n=1$: credit-card). We expand it with the Bank Account Fraud suite [12] (Feedzai, NeurIPS 2022): six bias-controlled variants (Base, I–V) of a large tabular fraud problem, a second fraud source, lifting the family to $n=7$. The result is an honest *negative/boundary* finding. On BAF the entire detector zoo is near chance (mean AUROC ≈ 0.6 ; BAF is engineered to be hard, with label noise and temporal drift), and there stacking is *not* significantly better than auto-select (+0.011, CI [−0.066, +0.064]) and sits *below* the best fixed detector (−0.033, CI [−0.098, +0.022]); on Variant I the per-task logistic stack even collapses below chance (0.375). This refines the central claim: the significant stacking advantage holds where base detectors are *individually informative*, and not on near-chance problems — combination amplifies signal that exists but cannot conjure it where the zoo is uniformly weak. Together with the time-series boundary, this confines the significant win to families with informative base detectors, rather than to “feature-vector” families per se (BAF is itself tabular).

Sealed external replication (D24). The strongest guard against an over-fit development split is a *sealed* holdout: the method, baselines, detector zoo, and pass condition were frozen (research_lock/SEALED_EXTERNAL_PROTOCOL_v1) *before* any holdout data was loaded, then scored exactly once with no post-hoc tuning. The holdout is 74 ADBench tasks under a *different feature extractor* than development (CV_by_ViT and NLP_by_RoBERTa vs. the ResNet18/BERT features used to build the 123-task archive), so these representations were never inspected. On this one-shot test (Table 31) the stack still beats auto-select (+0.016, CI [+0.004, +0.032]) and the best fixed detector (+0.043, CI [+0.022, +0.073]); both CIs exclude zero. The effect therefore *replicates*

under representation shift, lifting the stacking result from development evidence to sealed evidence (Gate U point 7). Honest scope: this is a sealed held-out under shifted representation on the same ADBench suite, not a wholly independent data source — that gap is closed next (D27).

Table 31: Sealed external holdout (D24, one-shot): method frozen before scoring the same 74 ADBench datasets re-featurized with an untouched ViT/roBERTa extractor (representation-shift holdout). Stacking beats both auto-select and the best fixed detector; both CIs exclude zero.

Method	mean AUROC	Δ (95% CI)
Best fixed detector (KNN)	0.768	—
Rank-mean	0.762	—
Simple average	0.764	—
Auto-select	0.795	—
Stack (logistic)	0.811	vs auto +0.016 [+0.004, +0.032] vs best +0.043 [+0.022, +0.073]

Fully independent external corpus (D27). The sealed holdout (D24) reuses the ADBench *suite* under new extractors; the hardest test is a different data *source* entirely. We froze the method and the exact 20-task list (`research_lock/INDEPENDENT_EXTERNAL_PROTOCOL_v1`) over sources *verifiably absent* from the 135-task archive — scikit-learn `digits` (10 one-vs-rest tasks), `wine` (3), `wdbc` (malignant = anomaly), and HAR smartphones (561 features, 6 activities) — and scored once with the construction “each class = anomaly, downsampled to 10%”. The result is the strongest replication in this thesis: stacking reaches mean AUROC 0.799 versus auto-select 0.689 (+0.111, CI [+0.052, +0.176]) and the best fixed detector 0.590 (+0.209, CI [+0.120, +0.303]); both CIs exclude zero (Table 32). The advantage is *larger* on these independent sources than on the development split, because the one-vs-rest tasks have informative detectors and single-detector selection is weak — exactly the regime where learned combination should and does win. This lifts the stacking claim from sealed evidence to *fully independent external* evidence.

Table 32: Fully independent external corpus (D27, one-shot): method and 20-task list frozen before scoring sources absent from the development archive (sklearn digits/wine/wdbc + HAR). Stacking beats auto-select and the best fixed detector; both CIs exclude zero.

Method	mean AUROC	Δ (95% CI)
Best fixed detector (COPOD)	0.590	—
Rank-mean	0.596	—
Simple average	0.598	—
Auto-select	0.689	—
Stack (logistic)	0.799	vs auto +0.111 [+0.052, +0.176] vs best +0.209 [+0.120, +0.303]

20 Multimodal Independent-Modality Failure (D23)

Every regime where reliability routing was inert used *single-input* families: all detectors see the same data, so no modality can fail independently. We therefore pre-registered (`research_lock/MULTIMODAL_RELIABILITY`) a test on true multimodal data (Real-IAD-D3: RGB, photometric-stereo, point-cloud XYZ experts), failing the deployment-best modality at test while the others stay clean; reliability routing detects the failed modality (KS validation-vs-test drift, no test labels) and routes away from it. All three

Table 33: D23 multimodal reliability on *two* real datasets (Real-IAD-D3, 15 categories; MVTec-3D, 7 categories). Mean AUROC in the modality-failure regime; the gate is inert in the clean regime (equals validation-only fusion) on both. Hypotheses are paired bootstrap over categories.

Method (failure regime)	Real-IAD-D3	MVTec-3D
Equal-weight (CW)	0.526	0.586
Stale auto-select	0.381	0.475
Validation-only fusion	0.519	0.581
Reliability gate	0.767	0.792
<i>Real-IAD-D3</i> : H1 vs equal-weight +0.241 [0.152, 0.348]; H2 vs stale-select +0.386 [0.281, 0.483]; H3 vs validation-only +0.248 [0.168, 0.323]. <i>MVTec-3D</i> : H1 +0.207 [0.172, 0.242]; H2 +0.317 [0.240, 0.405]; H3 +0.211 [0.180, 0.244]. All 95% CIs exclude zero.		

pre-registered hypotheses pass (Table 33); in the clean regime the gate is inert. The precise claim: *reliability routing helps iff modalities can fail independently*.

Failure-model dependence and real degradation (honest reality check). The D23 wins use the canonical *uniform-noise* modality failure (the failed sensor emits garbage, which *drifts* detectably). Two checks with more realistic faults locate the gate’s true regime. **(i) Real sensor dropout** (MulSen-AD, stuck/constant reading): the gate still beats stale single-modality selection (+0.28) but *not* plain fusion (vs equal-weight -0.03 , n.s.; vs validation-only fusion -0.05). **(ii) Real natural degradation** (D29: Real-IAD-D3 per-modality deep-PatchCore scores from the raw 259 GB, 16 categories, no injection): the gate beats stale selection (+0.13) but not plain fusion (+0.02/+0.03, n.s.). The honest refinement: *fusion robustly beats selection, but the reliability gate’s extra value over fusion is specific to drift-inducing faults*; under a stuck sensor or mild natural degradation there is no test-time drift to exploit and plain fusion is as good — exactly the inertness proposition.

A new method and a conjecture (per-sample routing). The inertness result is specific to *per-channel* reweighting (constant across samples). This motivates a genuinely new combiner: *per-sample reliability-gated stacking* (PS-RGS), whose weights $w_m(x)$ vary with the input via no-label local-reliability features and thus lie *outside* the linear class the inertness proposition bounds. We conjecture (T10) that per-sample gating strictly exceeds the best linear stacker *iff* per-sample reliability is label-relevant and heterogeneous across the input space. On the 123-task suite PS-RGS gives a small directional improvement over plain stacking (+0.0068, CI $[-0.0002, +0.015]$; 83/123 wins) that does *not* reach significance, and its per-task gain correlates positively but not significantly with detector heterogeneity (Spearman $r \approx 0.14$, $p \approx 0.12$) — the right direction for T10, but underpowered on a homogeneous classical zoo. PS-RGS is therefore a theoretically-grounded direction with honest *preliminary* evidence, not a confirmed win.

We then *ran* the natural next experiment — PS-RGS on a heterogeneous deep+classical zoo (adding DeepSVDD, an autoencoder, and a VAE) — and pre-registered a one-shot held-out confirmation (protocol D32) to guard against a development false-positive. The result is an instructive **honest negative**. On tabular *development* data the deep zoo made PS-RGS significantly beat plain stacking (+0.016, CI excl. 0); but on the *pre-registered held-out* families the effect *did not replicate*: ADBench-CV +0.002 (n.s.) and ADBench-NLP -0.020 (significantly *worse*). The tabular significance was therefore a development-set artefact, and the pre-registered one-shot caught it. PS-RGS is thus **not** a confirmed improvement; we report it as a theoretically-motivated method refuted by held-out confirmation — a textbook illustration of why pre-registration is the backbone

of this thesis. The only robust effect from the deep zoo is minor: it modestly improves *plain* stacking on feature-vector families (tabular +0.010, CV +0.004; both CIs exclude 0; n.s. on NLP). No SOTA or breakthrough claim follows: the absolute levels (0.87/0.78/0.65 across tabular/CV/NLP) are not leaderboard-beating, and the new method does not even reliably beat plain stacking out-of-family.

21 Conclusion

ELARA-U is now best framed as a universal anomaly meta-routing thesis rather than a clean-transfer superiority paper. The current result is promising: rank-normalized stacking substantially improves average rank and regret under a development stress benchmark where validation metrics become stale. The honest scientific boundary is equally clear: the current evidence is not enough for a sealed universal SOTA claim, and the reliability/drift novelty must be strengthened beyond plain stacking. The next phase should therefore prioritize natural shift, sealed holdouts, stronger ablations, calibrated abstention, and a clean reproducibility package.

A Theoretical Proofs and Extensions

A.1 Supporting Propositions and Lemmas

Proposition 3 (Rank-Normalized Stacking Invariance). *Let detector m produce scores $s_m(x)$, and let $h_m : \mathbb{R} \rightarrow \mathbb{R}$ be any strictly increasing transformation. Define the normalized rank feature*

$$r_m(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{s_m(x_i) \leq s_m(x)\}.$$

Then replacing $s_m(x)$ with $h_m(s_m(x))$ leaves $r_m(x)$ unchanged for every sample x , up to tie-handling. Therefore, any ELARA-U stacker using only detector-wise rank-normalized features is invariant to monotone score rescaling, calibration drift, or sensor-scale drift.

Proof. For any two samples (x_i, x_j) ,

$$s_m(x_i) \leq s_m(x_j)$$

if and only if

$$h_m(s_m(x_i)) \leq h_m(s_m(x_j)),$$

because h_m is strictly increasing. Therefore, the ordering of all samples under detector m is unchanged. Since $r_m(x)$ depends only on the ordering of scores and not on their numerical scale, $r_m(x)$ is unchanged. Any downstream stacker that receives only $r_m(x)$ is therefore invariant to such monotone transformations. $\square$

Interpretation. This proposition justifies the rank-normalized stacking design. It protects the router from meaningless score-scale changes while preserving anomaly ranking information.

Proposition 4 (Degenerate-Channel Guard for Saturation). *Let a fused anomaly score be*

$$g(x) = \sum_{m=1}^M w_m s_m(x), \quad w_m \geq 0, \quad \sum_m w_m = 1.$$

If detector k is saturated on validation data so that

$$\text{Var}_V(s_k) \leq \epsilon,$$

then replacing w_k by 0 and renormalizing the remaining weights changes pairwise score differences by at most $O(\sqrt{\epsilon})$. In the limit $\epsilon \rightarrow 0$, the saturated detector contributes no ranking information, and removing it cannot remove meaningful rank signal.

Proof. For any two samples (x_i, x_j) ,

$$|s_k(x_i) - s_k(x_j)| \leq O(\sqrt{\epsilon})$$

by the bounded-variance assumption. The contribution of detector k to the fused pairwise difference is

$$w_k(s_k(x_i) - s_k(x_j)),$$

which is also $O(\sqrt{\epsilon})$. Therefore, as $\epsilon \rightarrow 0$, detector k 's contribution to every pairwise ranking comparison vanishes. Removing k and renormalizing the remaining non-degenerate detectors does not remove meaningful rank information from the fused score. $\square$

Interpretation. This supports the degenerate-channel guard for saturated detectors. A detector that produces nearly constant scores should not influence routing or fusion.

Proposition 5 (Inverted Detector Harm Under Positive Fusion). *Assume the clean fused margin between an anomalous and normal sample is*

$$D_0 = g_0(X^+) - g_0(X^-),$$

with $\mathbb{E}[D_0] > 0$. Suppose an inverted detector k has margin

$$D_k = s_k(X^+) - s_k(X^-)$$

with $\mathbb{E}[D_k] < 0$. For a positive fusion weight $w > 0$, the fused margin

$$D_w = (1 - w)D_0 + wD_k$$

satisfies

$$\mathbb{E}[D_w] = (1 - w)\mathbb{E}[D_0] + w\mathbb{E}[D_k] < \mathbb{E}[D_0].$$

Under a Gaussian equal-variance margin model, AUROC is monotone increasing in the standardized margin mean. Therefore, assigning positive weight to an inverted detector decreases AUROC relative to zeroing it.

Proof. The expectation follows from linearity:

$$\mathbb{E}[D_w] = (1 - w)\mathbb{E}[D_0] + w\mathbb{E}[D_k].$$

Since $\mathbb{E}[D_k] < 0 < \mathbb{E}[D_0]$, we have $\mathbb{E}[D_w] < \mathbb{E}[D_0]$. Under the Gaussian equal-variance margin model,

$$D_w \sim \mathcal{N}(\mu_w, \sigma^2),$$

and

$$\text{AUROC}(g_w) = \Pr[D_w > 0] = \Phi\left(\frac{\mu_w}{\sigma}\right).$$

Because Φ is strictly increasing and $\mu_w < \mu_0$, positive weight on the inverted detector decreases AUROC. $\square$

Interpretation. This proposition supports validation-based inversion guards. If a detector is reliably below chance, positive-weight fusion can harm ranking.

Proposition 6 (Meta-Router Generalization Requires Enough Tasks). *Let $\mathcal{H}$ be a finite class of meta-routers, and let the task-level loss $\ell(h, \tau) \in [0, 1]$ denote normalized rank regret or negative-transfer loss. Given T independent meta-training tasks, with probability at least $1 - \delta$, every router $h \in \mathcal{H}$ satisfies*

$$L(h) \leq \hat{L}_T(h) + \sqrt{\frac{\log |\mathcal{H}| + \log(1/\delta)}{2T}},$$

where $L(h) = \mathbb{E}_\tau[\ell(h, \tau)]$ is the expected held-out task loss and $\hat{L}_T(h) = \frac{1}{T} \sum_{t=1}^T \ell(h, \tau_t)$ is the empirical meta-training loss.

Proof. For any fixed h , Hoeffding’s inequality gives

$$\Pr \left(L(h) - \hat{L}_T(h) > \epsilon \right) \leq \exp(-2T\epsilon^2).$$

Applying a union bound over all $h \in \mathcal{H}$,

$$\Pr \left(\exists h \in \mathcal{H} : L(h) - \hat{L}_T(h) > \epsilon \right) \leq |\mathcal{H}| \exp(-2T\epsilon^2).$$

Set the right-hand side equal to δ :

$$|\mathcal{H}| \exp(-2T\epsilon^2) = \delta.$$

Solving for ϵ gives

$$\epsilon = \sqrt{\frac{\log |\mathcal{H}| + \log(1/\delta)}{2T}}.$$

Thus, with probability at least $1 - \delta$, the stated bound holds for every $h \in \mathcal{H}$. $\square$

Interpretation. This explains why ELARA-U needs many tasks and leave-family-out evaluation. A high-capacity router trained on too few tasks can overfit meta-level structure and fail to generalize.

A.2 Old ELARA/RGA Source-System Theory Stack

Below are the foundation certificates and bounds inherited from the ELARA/RGA source-system framework.

Certificate 1 (Switching Certificate). *Let $\Delta_{\text{fired}} = A(\text{route}) - A(\text{default})$ be the routed-minus-default performance on the subset of inputs where the router fires. The router may replace the default only if the one-sided level- α lower confidence bound satisfies $\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0$. Under this rule the expected switch is non-negative at level α : routing is admitted only with certified fired-set benefit, never on a point estimate.*

Diagnostic 1 (Mean-Reliability Dilution). *Let per-expert reliabilities be $r_1, \dots, r_M$ with mean $\bar{r} = \frac{1}{M} \sum_m r_m$ and gate threshold τ . There exist configurations with $\bar{r} > \tau$ while a minority expert collapses ($r_k \rightarrow 0$): mean-gating does not fire and the collapsed expert is still trusted. The miss probability is $P(\text{miss}) = \Phi((\bar{\mu} - \tau)/\bar{\sigma})$; detecting minority collapse requires per-expert / min-gating, not the mean (Theorem T3).*

Proposition 1 (Router Generalisation Slack). *Let the router class have complexity C (e.g. VC dimension or Rademacher complexity) and let the meta-training set contain T tasks. The gap between meta-training and held-out expected rank-gain is bounded by $O(\sqrt{C/T})$ with high probability. Hence the evidential strength of a routing claim requires $T \gtrsim C$: a high-capacity router over few tasks yields weak evidence even at a positive point estimate. This motivates a large, heterogeneous benchmark suite (the Gate U minimum-viable families) before any claim.*

References

- [1] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” in *SIGMOD*, 2000.
- [2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *ICDM*, 2008.
- [3] L. Bergman and Y. Hoshen, “Classification-based anomaly detection for general data,” in *ICLR*, 2020.
- [4] K. Roth et al., “Towards total recall in industrial anomaly detection,” in *CVPR*, 2022.
- [5] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *ICLR*, 2017.
- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *ICML*, 2017.
- [7] Y. Zhao, R. Rossi, and L. Akoglu, “Automating outlier detection via meta-learning,” in *NeurIPS*, 2021.
- [8] S. Han, X. Hu, H. Huang, M. Jiang, and Y. Zhao, “ADBench: Anomaly detection benchmark,” in *NeurIPS Datasets and Benchmarks*, 2022.
- [9] Y. Zhao, Z. Nasrullah, and Z. Li, “PyOD: A Python toolbox for scalable outlier detection,” *JMLR*, vol. 20, 2019.
- [10] D. H. Wolpert, “Stacked generalization,” *Neural Networks*, vol. 5, no. 2, 1992.
- [11] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *NeurIPS*, 2017.
- [12] S. Jesus, J. Pombal, D. Alves, et al., “Turning the tables: Biased, imbalanced, dynamic tabular datasets for ML evaluation (Bank Account Fraud),” in *NeurIPS Datasets and Benchmarks*, 2022.